

RISHU BAGGA

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EDUCATION

University of Illinois, Urbana - Champaign
B.S Computer Science

August 2017 - May 2021

EXPERIENCE

The San Francisco Compute Company - Cloud Team
Software Engineer

March 2026 - June 2026
San Francisco, CA

- Reduced postgres db footprint 99.99% by offloading historical machine heartbeat data to AWS S3 via AWS Firehose.
- Implemented a managed secrets service for customers to store sensitive information without baking it into their VM image or codebase.
- Implemented an EC2 IMDSv2 style metadata service, and rate limiter to run on every machine in our fleet.
- Implemented Instance Priorities: a feature to allow a customer to customize which instances get preference in allocation and preemption. <https://docs.sfccompute.com/preview/instances#instance-priority>
- Implemented end to end integration tests in the company's CI pipeline to eliminate 99% of unpredictable breaking changes across different services of the product.
- Integrated permissions systems to enable RBAC for all API endpoints on our cloud offering.

Amazon Web Services - Aurora PostgreSQL Engine Team
Software Development Engineer II

July 2021 - December 2024
Seattle, WA

- Facilitated the release of Aurora Postgres v14 - revising 100+ upstream community Postgres changes to be compatible with proprietary Aurora Storage system.
- Submitted patches to the open source postgres community on behalf of AWS, for moving SLRU cached components into the main buffer cache, adding page headers to the SLRU cached components, and revising the upgrade codepath to support the changes.
- Assisted in drafting innovative designs for ultra fast major version upgrades of Aurora Postgres.
- Designed throughput improvements for the Async / Notification subsystem in Aurora Postgres, resulting in over 30x performance gains.
- Handled 100+ real time production issues to prevent customer downtime on production databases.
- Analyzed root cause for 100+ customer requests and SLA violations

Amazon Web Services (AWS)
Software Development Engineering Intern

May 2020 - August 2020
Palo Alto, CA

- Helped identify blind spots in the Aurora service by building an internal metrics aggregating tool built with S3, Redshift, and Grafana. (Java)

SimBioSys Inc.
Software Engineering Intern

May 2019 - August 2019
Champaign, IL

- Built website for Tumorscope product, product using Docker, Flask, TypeScript
- Helped train breast tumor recognition model by generating ground truth data for 100+ MRI images

SIDE PROJECTS / OTHER WORK

Personal Blog
rishuwall.vercel.app

- deployed using Supabase, Vercel, and Railway
- used APIs from Strava, and Spotify

Postgres Community Involvement

- Gave a talk on the Postgres SLRU subsystem at Seattle Postgres Users Group: https://www.youtube.com/watch?v=1ucqA8yy_RM
- Worked on and contributed to discussion on a series of patches to improve the LISTEN/NOTIFY system in Postgres. Some suggestions were merged into core. <https://github.com/postgres/postgres/commit/282b1cde9dedf456ecf02eb27caf086023a7bb71>